

Jiban Chaudhary

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Personal

- Date of Birth: May 17, 2001.
- Language: Nepali, English, Hindi
- Gender: Male
- Hobbies: Football, Swimming, Trekking

Education

Advanced College of Engineering and Management

Kathmandu, Nepal

B.E. in Computer Engineering. GPA: 73.23%

June 2025

- Undergraduate coursework in Data Structures & Algorithms, Discrete Mathematics, Operating Systems, Databases, and Machine Learning.

Xavier International College

Kathmandu, Nepal

High School (Science). GPA: 3.21

July 2019

Research

ACEM Computer Engineering

Kathmandu, Nepal

Experience

Legal Advisor using NLP and Transformer

and Projects

Supervisor: Assoc. Prof. Dr. Nanda Bikram Adhikari ([Google Scholar](#))

My research focuses on the design and development of an intelligent legal advisory system for cybercrime using state-of-the-art Natural Language Processing (NLP) and Transformer-based models. The work addresses limitations of traditional rule-based legal systems by incorporating contextual understanding, semantic retrieval, and automated summarization to improve legal accessibility and decision support.

- Designed an end-to-end NLP pipeline for cybercrime legal queries, incorporating query preprocessing, validation, and relevance detection. A LangChain-based agent was used to distinguish legal from non-legal inputs, ensuring robust handling of user queries and reducing irrelevant query processing.
- Fine-tuned a Legal-BERT model and implemented a Retrieval-Augmented Generation (RAG) framework using sentence-transformer embeddings to classify cybercrime-related queries and retrieve relevant statutes, policies, and incident reports from a curated legal dataset, enabling context-aware legal information retrieval beyond keyword matching.
- Developed a privacy-aware and scalable legal advisory chatbot following Agile methodology and compliant with the Electronic Transactions Act (ETA) 2063, generating structured and summarized legal guidance to improve accessibility and interpretability of cybercrime-related legal information for non-expert users.

Deepfake Detection using XceptionNet and LSTM

- Developed a deep learning-based deepfake detection system by combining a pretrained XceptionNet backbone for spatial feature extraction with an LSTM

module for temporal modeling across video frames, enabling robust classification of real vs manipulated videos.

- Implemented video preprocessing modules to extract and normalize frames, and built an inference pipeline that classifies videos in real time with low latency.
- Achieved overall model accuracy of ~86% on test data, demonstrating practical effectiveness of hybrid CNN–LSTM architectures for sequential video analysis.

Professional Experience

O.Labs(Olive Group) – Kathmandu, Nepal

Data Engineer

June 2025 – Present

- **Designed and implemented end-to-end data pipelines** for machine learning workflows, covering raw data ingestion, preprocessing, multi-model inference, post-processing, and structured dataset generation.
- **Integrated multiple machine learning and computer vision models** within unified pipelines, including transformer-based models, segmentation architectures (SAM-style models), object detection, OCR, and super-resolution models, ensuring consistent data flow across heterogeneous architectures.
- **Developed data transformation and intermediate representation layers**, converting between images, tensors, NumPy arrays, and tabular formats, and resolving input/output constraints such as channel formats, tensor shapes, and datatype compatibility.
- **Analyzed and stabilized ML data pipelines** by debugging dependency conflicts, framework version mismatches, duplicate outputs and error in data-model interfaces.
- **Conducted applied research and experimentation** on model architectures and pipeline design, evaluating base vision architectures, transformer models, and segmentation approaches to study the effects of preprocessing strategies, inference configurations, and integration choices on pipeline reliability and output quality.

Data Engineer Intern

Feb 2025 – June 2025

- Gained hands-on experience in data preprocessing, feature engineering, and ML pipeline optimization for real-world projects.
- Conducted research on neural networks, transformers, and PCA, applying mathematical concepts with NumPy and Pandas for data transformation and analysis.
- Collaborated on team projects, demonstrating problem-solving, analytical thinking, and contribution to AI-driven solutions.

CloudFactory– Kathmandu, Nepal

Data Specialist

Dec 2019 – July 2022

- Specialized in visual data processing, preparing structured datasets for machine learning applications.
- Enhanced data annotation and analysis workflows using Excel and Python, supporting model training and evaluation.
- Contributed to AI projects by transforming raw unstructured data into organized, structured information.

Technical Skills

- **Programming & Data Analysis:** Python, SQL, C++, NumPy, Pandas, Matplotlib, Excel

- **Machine Learning & AI:** TensorFlow, PyTorch, Transformers, CNNs, LSTMs, PCA, Computer Vision, Deep Learning Models
- **Data Engineering & Research Tools:** Data preprocessing, pipeline design, feature engineering, model optimization
- **Web & Database:** Django, MySQL, SQLite
- **Scientific Computing & Visualization:** Jupyter Notebook, Google Colab, Matplotlib, Seaborn.